HELORS 2026

HELLENIC OPERATIONAL
RESEARCH SOCIETY

11th International Symposium & 33rd National Conference on Operational Research

Operational Research for Sustainable Development

XANTHI, GREECE
29—31 OCTOBER 2026

ORGANIZED BY

HELORS HELLENIC OPERATIONAL RESEARCH SOCIETY

ΔΗΜΟΚΡΙΤΕΙΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΡΑΚΗΣ | DEMOCRITUS UNIVERSITY OF THRACE

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HELORS 2026 Conference Program

Conference of the Hellenic Operational Research Society · 78 presentations in 17 sessions

29–31 October 2026 · Xanthi, Greece

Department of Production & Management Engineering, Democritus University of Thrace · helors2026.eu

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WELCOME · Thu 29 · 9:00-10:00 · Auditorium

Registration & Welcome Coffee

SESSION 01 · Thu 29 · 10:00-12:00 · Room A1

Climate Adaptation and Environmental Decision Making

Chair: Symeon Symeonidis

Wildfire Risk Mapping and Optimal Deployment of Monitoring Systems Using GIS-Based Analysis with MCDA Methods and NSGA-II Algorithm
Apostolia-Theodora Drakopoulou, Stella-Maria Antoniadi, Vasilis Alexandridis, Stergios Diamantopoulos, Apostolia Papadoudi

Integrating Nature-Based Solutions into Graphical Digital Twins for Climate Adaptation Decision Support
Parthena Ioannidou, Symeon Symeonidis, Vasilis Perifanis, Christos Akratos, Antonios Gasteratos

Multi-Criteria Decision Support in Climate Policy: A Decade-Long Systematic Literature Review Following the Paris Agreement
Avgi Rousogiannaki, Konstantinos Koasidis, Georgios Trachanas, Haris Doukas, John Psarras, Georgios Xexakis, Alexandros Nikas

Effect of the Preference Function on Net Flow Discrimination in PROMETHEE II
Thomas Bakas, Mike Spiliotis

Utilizing System Dynamics and System Archetypes to Examine Interrelations Among the Factors of the Environmental Footprint
Konstantina Theona, Athanasios Spyridakos, Nikolaos Tsotsolas

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SESSION 02 · Thu 29 · 10:00-12:00 · Room A2

AI, Technology Adoption and Digital Transformation

Chair: Fotis Kitsios

What Drives Human-Robot Collaboration? An Empirical Study of Industrial Employees
Vasilidou Maria, Kalaitzopoulos Ioannis, Diamantidis Anastasios

Beyond Adoption: Verification Behavior and Undisclosed Generative AI Use in the Workplace
Dora Katsavou, Georgios Theriou, Dimitrios Chatzoudes, Konstantinos Tasoulis, Foteini Kravariti, Maria Progoulaki

IT Capabilities and Emerging Technology Exploitation
Ioannis Sitaridis, Fotis Kitsios

A Multi-Criteria Decision Analysis Approach to Digital Maturity Assessment: Addressing Methodological Rigidity
Mitroulis Dimitrios, Kitsios Fotis

What Drives ERP Satisfaction in SMEs? Comparing Usability, User Experience and Support
Konstantina Remidianaki, Christos Ziakis

CEREMONY · Thu 29 · 12:00-12:30 · Auditorium

Opening Ceremony

AWARD · Thu 29 · 12:30-12:45 · Auditorium

HELORS National Award

PLENARY TALK · Thu 29 · 12:45-13:45 · Auditorium

Prof. Alexis Tsoukias

Designing alternatives for decision support: A new challenge

Light Lunch Break · Thu 29 · 13:45-14:30

SESSION 03 · Thu 29 · 14:30-16:30 · Room A1

Facility Location and Network Design

Chair: Hercules Dimitriadis

Enhancing Facility Reliability with Backup Options and Budget Constraint
Abdulrahman Alenezi

Spatial and Siting Analysis of Sustainable Biomass Baseload Power Generation for Nigeria: OR for Sustainable Development
Adedolapo Oluyomi Akinde

EV Charging Network Optimisation: Integrating Battery Swapping with (s, Q) Inventory Policy
Debapriya Basu, KM Rashmi, Prasun Das

Mobility-Aware Electric Vehicle Charging Infrastructure Planning: A Case Study of Kozani, Greece
Hercules Dimitriadis, Dimos Tsouros, Nikolaos Ploskas

Multistage Stochastic Mobile Healthcare Facility Location Under Facility Disruption Uncertainty
Bonn Kleiford Seranilla

SESSION 04 · Thu 29 · 14:30-16:30 · Room A2

Consumer Behaviour, Satisfaction and Pricing

Chair: Georgia Parastatidou

Mapping Competitive Performance: A Hybrid Framework for Multi-Store Retail Networks
Anastasia S. Saridou, Athanasios P. Vavatsikos

Locality as a Digital Communication Tool: From Online Visibility to Consumer Response
Vassiliki Aggelidi, Georgia Parastatidou, Vassilios Chatzis

Perception, Intention, and Self-Reported Actual Spending as a Basis for Strategic Local Branding
Paraskevi Bakou, Georgia Parastatidou, Vassilios Chatzis

When the Locally Optimal Decision Degrades the System: Field Evidence from a Live Marketplace Pricing Operation
Elias Sultan

MUSAsent: Integrating Aspect-Based Sentiment into Multicriteria Satisfaction Analysis
Anastasios Kyriakidis, Stelios Tsafarakis

SESSION 05 · Thu 29 · 14:30-16:30 · Auditorium

Project Scheduling and Metaheuristics

Chair: Georgios K. Koulinas

A Whale Optimization Algorithm for the Classic Resource-Constrained Project Scheduling Problem
Nikolaos T. Mitsou, Georgios K. Koulinas

A Constrained Grey-Box Optimization Method with Swarm-Guided Acquisition and Multi-Anchor Boundary Repair
Emmanouil Karantoumanis, Nikolaos Ploskas

A Sequential Feasibility and Criticality Assessment Framework for Engineer-to-Order Rocket Development Projects
Ecem Sıla Aşıcı, Müjde Genevois

A Coronavirus Disease Optimization Algorithm for Project Resource Scheduling
Iordanis Iordanidis, Nikolaos Mitsou, Georgios Koulinas

Coffee Break · Thu 29 · 16:30-17:00

SESSION 06 · Thu 29 · 17:00-18:30 · Room A1

Advances in Multicriteria and Hybrid Decision Analysis

Chair: Pavlos Delias

Sustainability Evaluation of Financial Institutions Using Multi-Criteria Decision-Making Methods
Georgios E. Tsakidis, Konstantinos I. Efthymiadis, Anastasia S. Saridou, Athanasios P. Vavatsikos

simOutrank: An R Package for Non-Compensatory Clustering
Pavlos Delias

Fuzzy MCDA for Circular Economy and Sustainability Monitoring in Rural Agrifood SMEs: The CERAS Experience
Vassiliki Kazana, Vasilios Plakandaras, Dimitrios Raptis

Hybrid AI-OR Decision Intelligence: Integrating Cloud and Local Agents with Machine Learning and Optimization for Sustainable Business Decisions
John Isaaris

SESSION 07 · Thu 29 · 17:00-18:30 · Room A2

Maritime and Air Transport Operations

Chair: Dimitrios Dimitriou

Integrated Evaluation Framework for Passenger Experience and Service Quality in Mediterranean Smart Cruise Ports
Dimitrios Dimitriou, Maria Sartzetaki, Aristi Karagkouni, Antonia Moutzouri

Data-Driven Assessment of Air Transport Network Resilience and Connectivity in the Mediterranean Region
Maria Sartzetaki, Iliana Kolari, Aristi Karagkouni, Dimitrios Dimitriou

Multi-Period Ship Technology Transition Planning Under Overlapping Carbon Regulations: A Bi-Objective Mixed-Integer Programming Approach
Simay Budak, M. Ömer Budak, A. Burak Paç, Ayşenur Budak

Geospatial Risk Assessment of Digital Ship Systems: A Systematic Review Towards an Interactive Mapping Tool for Decision Support
Panagiota Rempi, Eugenia Skepetari, Anastasios Safaris, George Papagiannakis, Christos Ntanos, Dimitris Askounis

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SOCIAL PROGRAM · Thu 29 · 20:30

Conference Gala Dinner

SESSION 08 · Fri 30 · 9:00-11:00 · Room A1

Mathematical Programming and Exact Methods

Chair: Argyrios Anagnostopoulos

Passenger-Weighted Optimization of Airport Slot Allocation Under Alternative Grandfather Rule Thresholds
Konstantia Kontodimou, Chris Mitsis, Konstantinos Kepaptsoglou

Benchmarking Exact Approaches on Variants of the Nurse Rostering Problem
Myrto Kolia, Christos Valouxis, Dimitrios Kiosses, Christos Gogos

Optimal Control of a Damaging Population Through Total Catastrophes
Epaminondas G. Kyriakidis

From Fixed Rosters to Demand-Adaptive Scheduling: A Rolling-Horizon MILP for Mobile Thermal Energy Storage Under Dynamic Occupancy and Forecast Uncertainty
Iordanis Iordanidis, Georgios Gaidajis, Argyrios Anagnostopoulos

Linearizing the Inverse and MinDistance Global Constraints
Filanthi Papadimitriou, Dimosthenis Tsouros, Nikolaos Ploskas

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SESSION 09 · Fri 30 · 9:00-11:00 · Room A2

Sustainable Urban Mobility and Transport Resilience

Chair: Aristi Karagkouni

Identifying Critical Urban Road Links Under Rainfall Disruption Scenarios in Thessaloniki
Dimos Touloumidis, Michael Madas

Integrating Sustainability and Climate Resilience in Urban Transport Decision Support: An Operational Research Perspective
Michael Madas, Alexandros Deligiannis, Dimos Touloumidis

A Multi-Criteria Decision Analysis Framework for Evaluating Business Models in Smart and Meta-City Ecosystems
Antonios Paraskevas, Ioanna Giannouli, Maria Votta, Michael Madas

Business Intelligence Modelling for Sustainability Appraisal in Transport Infrastructure
Aristi Karagkouni, Vasileios Benteniotis, Maria Sartzetaki, Dimitrios Dimitriou

Smart Infrastructure Project Management for Green and Resilient Urban Mobility
Christos Papadopoulos, Dimitrios Dimitriou

AWARD · Fri 30 · 11:00-11:10 · Auditorium

HELORS Award

PLENARY TALK · Fri 30 · 11:10-12:10 · Auditorium

Prof. Harris Doukas

Decision-Making for Sustainable Cities: Lessons from Athens

Coffee Break · Fri 30 · 12:10-12:30

SESSION 10 · Fri 30 · 12:30-14:30 · Room A1

Energy Communities, Renewables and Spatial Planning

Chair: Pantelis Botsaris

Cross-Border Energy Communities in European Region
Charalampos Tsimpiriadis, Pantelis Botsaris, Athanasios Vavatsikos

The Impact of Establishing Energy Communities on Tourism in the Region of Eastern Macedonia and Thrace (REMTH)
Charalampos Tsimpiriadis, Pantelis Botsaris, Athanasios Vavatsikos

Digital Decision Support Platform for Residential Energy Efficiency Renovation One Stop Shop: Definition of User Roles and Interactions
Konstantinos Kefalas, Aikaterini Papapostolou, Ioanna Andreoulaki, Alexandra Soufleri, Vangelis Marinakis

From Solar Potential to Youth Leadership: A Spatial Multi-Criteria Approach to Targeting Community Energy Initiatives
Iason C. Dimitriou, Evi Makri, Dr. Apostolos Arsenopoulos, Vangelis Marinakis

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SESSION 11 · Fri 30 · 12:30-14:30 · Room A2

Energy Infrastructure, Resources and Circular Economy

Chair: Spiros Alexopoulos

Prioritizing Physical Asset Climate Resilience Investments in Gas Transmission Infrastructure: A MACBETH Multi-Criteria Framework
Spiros Alexopoulos

Performance Assessment of Circular Economy Across the EU: A Multi-Criteria Assessment
Georgios P. Trachanas, Konstantinos Kostas, Charalampos Papavasileiou, Anastasios Karamaneas, Konstantinos Koasidis, Dimitris Askounis, Alexandros Nikas

Operational Integration of Packed Bed Latent Heat Storage and Organic Rankine Cycle Systems for Waste Heat Recovery
Konstantinos Parissis, Argyrios Anagnostopoulos, George Gaidajis, Panos Seferlis

Contingency-Based AC Optimal Power Flow for Distributed Generation Planning with Critical Hospital Loads
Gamze Yetimler Kostur, Kemal Sarıca

SESSION 12 · Fri 30 · 12:30-14:30 · Auditorium

Vehicle Routing and Logistics Optimization

Chair: Georgios Papagiannakis

Air-Quality-Aware Urban Routing Under Changing Environmental Conditions: Assessing Route Recommendation Robustness in Thessaloniki
Alexandros Deligiannis, Michael Madas

A Dynamic Programming Algorithm for Choosing the Optimal Next Customer in a Stochastic Vehicle Routing Pproblem with Deliveries
Theodosis D. Dimitrakos, Epaminondas G. Kyriakidis, Andreas F. Papasalouros

A Unified Framework for Comparative Clustering Evaluation and Stability-Aware Fragility-Constrained Industrial Pallet Loading
Thomas Mantzou, Dimitrios Varsamis, Sotirios Dimitriadis

Design of an AI-Driven Decision Support Architecture for Real-Time Multi-Sensor Mass Spectrometry Monitoring of Cargo Conditions in Land Transportation
Georgios Papagiannakis, Konstantinos Tsokas, Sofia Karma, Sofianos Lampropoulos, Ariadni Michalitsi-Psarrou, Christos Ntanos

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Light Lunch Break · Fri 30 · 14:30-15:00

SESSION 13 · Fri 30 · 15:00-17:00 · Room A1

Health Systems and Medical Decision Support

Chair: Nikolaos Tsotsolas

Uncertainty-Aware Selective Classification of Brain Tumor MRI Using Fuzzy Confidence and Evolutionary Optimization
Natasa Milosavljevic, Nebojsa Ralevic

AI-Driven Clinical Decision Support Through Heterogeneous Data Integration and Dynamic Scientific Knowledge Retrieval
Emmanouil D. Gkagkalis, Leonidas L. Fragidis

Fuzzy and Stochastic Multi-Criteria Decision Aid Classification Methodological Framework for Guidelines and Protocols Recommendations in Cardiopulmonary Resuscitation and First Aids
Athanasios Spyridakos, Nikolaos Nikolaou, Nikolaos Tsotsolas, George Pierrakos

Strategic Blood Supply Coordination Planning in Conflict Settings
Claudia Franzè, Noura El Hassan, Bonn Kleiford Seranilla

A Generalized Decision System for Optimal Lockdown Timing During an Epidemic
Noura El Hassan, Bacel Maddah, Fouad Ben Abdelaziz

SESSION 14 · Fri 30 · 15:00-17:00 · Room A2

Efficiency and Performance Measurement

Chair: Christina Bampatsou

Constructing a Macro Quality Index (MQI): A Multi-Criteria Decision Analysis Approach
Pantelis Sotirelis, Michalis Doumpos

Efficiency Evaluation Considering Hierarchical Structure: An Application on the EU Agricultural Systems
Leonidas Sotirios Kyrgiakos, Siwei Xiao, Marios Dominikos Kremantzis, Georgios Kleftodimos, Marios Vasileiou, George Vlontzos

An Integrated DEA-Based Method for Assignment Problems Incorporating Multi-Attribute and Slack Measures
Shantnu Verma, Shivi Agarwal, Trilok Mathur

A SCOR-Based Decision Support System for Supply Chain Performance Measurement Using MCDM Methods
Georgios Sidiropoulos, Vasileios Zeimpekis, Nikos Tsotsolas

Efficiency Evaluation of Bakery Chain Using Data Envelopment Analysis: An Application of the Two-Stage DEA Model
Iordana A. Vangeli, Athanasios P. Vavatsikos

Coffee Break · Fri 30 · 17:00-17:30

SESSION 15 · Fri 30 · 17:30-19:30 · Room A1

Supply Chain Resilience, Skills and Workforce

Chair: Anastasios Diamantidis

A Decision-Oriented Capability-Conversion Framework for Robust and Resilient Manufacturing Supply Chains
Fadhlur Rahim Azmi, Nursyahwani Mohd Sukri, Veera Pandiyan Kaliani Sundram

Human Resource Management and Skills Development for Resilient Logistics Operation
Stylianos Lyras, Maria Sartzetaki, Dimitrios Dimitriou

Strategic Supply Chain Management and Firm Performance
Aikaterini Pantou, Dimitrios Prosmas, Anastasios Diamantidis, Prodromos Chatzoglou

Towards a Framework for Intervention Design in Food Security Systems Based on Spectral Analysis of DEMATEL Studies
Kerasia Kalkitsa, Pavlos Delias, Kalliopi F. Sotiropoulou

Identifying Emerging Skill Configurations Through Latent Class Analysis: A Framework for Future Skills Research
Ioannis Zervas, Emmanouil Stiakakis

SESSION 16 · Fri 30 · 17:30-19:30 · Room A2

Production Planning, Scheduling and Simulation

Chair: Alexandros Xanthopoulos

Big-Bucket, Single-Level, Single-Machine, Sequence-Dependent Capacitated Lot-Sizing Problem with Setup Carryover and Stochastic Demand: A Case Study
Nour Ramadan, Sana Ghannam

Simulation Analysis of an Unreliable Production System with Imperfect Product Quality
Eleni Katsika, Alexandros Xanthopoulos

Combined Pull Production Control and Job Scheduling in Discrete-Event Manufacturing Systems
Anastasia Karamanli, Alexandros Xanthopoulos

Assessing the Effect of Symmetry Breaking in the Distributed Permutation Flow-Shop Scheduling Problem
Kiosses Dimitrios, Christos Valouxis, Myrto Kolia, Christos Gogos

Spare Parts Optimization for Marine k-out-of-N Systems with Weibull Component Aging
Maria Papathemeli

SESSION 17 · Sat 31 · 9:00-11:00 · Room A1

OR and Emerging Technologies for Sustainable Engineering Systems

Chair: Athanasios P. Vavatsikos

Operations Research and Sustainable Mining
Georgios Gaidajis

Solving the CVRP Through Deep Reinforcement Learning: Theoretical Analysis of Decision Mechanisms and Computational Performance Evaluation
Evmenios D. Konstantinidis, Georgios K. Koulinas

Spatial Assessment of Wind Energy Potential in Greece
Maria T. Maroulasova, Anastasia S. Saridou, Athanasios P. Vavatsikos

A Deep Neural Network-Guided Local Search Algorithm for the Classic RCPSP
Nikolaos T. Mitsou, Georgios K. Koulinas

CLOSING · Sat 31 · 11:00-12:00 · Auditorium

Round Table-Closing Session

SOCIAL PROGRAM · Sat 31 · 12:00-14:00

Guided Tour to the Old Town of Xanthi

Sessions run in parallel in Room A1, Room A2 and the Conference Auditorium.

Presenting author in red